

Sidarth Kulkarni

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EDUCATION

Northeastern University

Bachelor of Science in Computer Science and Financial Technology, Dean's List

Boston, MA

Aug. 2021 – Dec. 2025

EXPERIENCE

Software Engineer Co-op

MORSE Corp

Jan. 2025 – June 2025

Cambridge, MA

- Designed and deployed runtime validation service for ML algorithms, raising compliance from **23%** to **95%**.
- Integrated YOLOv8 models into production inference pipeline, optimizing PyTorch operations to cut GPU memory usage by **76%**.
- Led 3-person agile team delivering AI-powered training data quality assessment tool using **Polars** in **66%** of allocated timeline.
- Engineered analytics pipeline in **Python** to compute and visualize novel object detection accuracy metrics.

Quantitative Analytics Intern

PNC Bank

May 2024 – Aug. 2024

New York City, NY

- Built Gradient Boosting Classifier (**0.81 AUC**) to predict consumer delinquency, presenting insights to C-suite executives.
- Developed feature engineering and selection pipeline with **PyCaret**, improving model AUC by **2%**.
- Architected SQL database and Python data pipeline for real-time investment tracking, reducing decision-making latency by **63%**.

Technical Project Management Intern

SiPhox Health

June 2023 – May 2024

Burlington, MA

- Built responsive web app with **React** frontend and **ExpressJS** backend to manage product orders end-to-end.
- Developed personalization quiz system using Typeform API, **Python**, and Zapier, generating **\$4K monthly revenue** with a **4%** conversion rate.
- Filed provisional patent for auxiliary blood collection device, engineered prototype in **AutoCAD**, and oversaw manufacturing with a **100%** sample collection success rate.

PROJECTS

Patient Health Assistant | Python, Flask, GCS, Docker, SQL, Git

Aug. 2024 – Present

- Developed AI-powered health Q&A system used by **15+ patients** and **10+ doctors** for clinical queries.
- Created FHIR-compliant **Flask** service integrating Epic Systems APIs with secure OAuth2 architecture.
- Fine-tuned Gemini model with semantic guardrails, reaching **91%** accuracy on benchmark medical questions.
- Deployed production backend handling 20+ monthly queries with <200ms latency.

RAG Benchmarking Tool | Python, FastAPI, ClickHouse, PostgreSQL, Docker, Streamlit

June 2025 – July 2025

- Built **RAG benchmarking system** to evaluate Vector Databases, integrating **ClickHouse** and **PostgreSQL** for scalable storage.
- Implemented microservices with Docker orchestration and async workers for deployment on NetApp's distributed clusters.
- Integrated LangChain + **Claude** to enable intelligent query-answering pipelines with full metrics collection in LangSmith.

TECHNICAL SKILLS

Languages: Python, Java, C++, SQL, TypeScript, JavaScript, C, VBA, Bash

Frameworks & Tools: PyTorch, TensorFlow, LangChain, Flask, FastAPI, React, Docker, Git, MongoDB, PostgreSQL, ClickHouse, scikit-learn

Platforms & Methodologies: AWS, GCS, Linux, Agile, CI/CD, MCP, PowerBI, Figma

INTERESTS

Electric Guitar, Tennis, 3D Modeling, UX Design, Running, Jazz Piano